

1. (a) (i) As A has two identical rows, $\det(A - 0 \cdot I) = \det(A) = 0$ and hence 0 is an eigenvalue of A . As the reduced row echelon of the augmented matrix $(A \mid 0)$ is

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right),$$

$$\begin{aligned} E_0 &= N(A) = \left\{ \begin{pmatrix} -r-s-t \\ r \\ s \\ t \end{pmatrix} : r, s, t \in \mathbb{R} \right\} = \left\{ r \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} : r, s, t \in \mathbb{R} \right\} \\ &= \text{Span}(\{u_1, u_2, u_3\}), \text{ where } u_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, u_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, u_3 = \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \end{aligned}$$

Moreover, $\{u_1, u_2, u_3\}$ is a linearly independent set, and hence a basis for E_0 .

(ii) As $A \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$, $\lambda^* = 4$ is an eigenvalue of A with the eigenvector $v = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$.

- (b) Set $Q = (v \ u_1 \ u_2 \ u_3) = \begin{pmatrix} 1 & -1 & -1 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$. Then Q is invertible because v, u_1, u_2, u_3 are linearly independent. Moreover,

$$AQ = (Av \ Au_1 \ Au_2 \ Au_3) = (4v \ 0 \ 0 \ 0) = Q \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

2. Note that the characteristic polynomial is $\det(A - tI_3) = (1 - t)(2 - t)(a - t)$.

- If $1 \neq a \neq 2$, then A has 3 distinct (real) eigenvalues, and hence A is diagonalizable.

- If $a = 1$, the algebraic multiplicity of $\lambda = 1$ is 2. As $A - I = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & b & 1 \end{pmatrix}$ has rank 1, its nullity is 2, and $\dim E_1 = 2$. This shows the geometric multiplicity of $\lambda = 1$ is 2. On the other hand, both algebraic and geometric multiplicities of $\lambda = 2$ are 1. Thus A is diagonalizable.

- If $a = 2$, the algebraic multiplicity of $\lambda = 2$ is 2. Note that $A - 2I = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & b & 0 \end{pmatrix}$.

If $b = 0$, then $A - 2I$ has rank 1 and nullity 2. As above, the geometric multiplicity of $\lambda = 2$ is also 2, and A is diagonalizable.

If $b \neq 0$, then $A - 2I$ has rank 2 and nullity 1. The geometric multiplicity of $\lambda = 2$ is 1, which is less than the algebraic multiplicity. So A is not diagonalizable.

To conclude, we must have $a = 2$ and $b \neq 0$.

3. (a) Consider $c_1v + c_2Av = 0$, then $A(c_1v + c_2Av) = A \cdot 0$ and hence

$$c_1Av = c_1Av + c_2A^2v = 0.$$

As $Av \neq 0$, $\underline{c_1 = 0}$ and hence $c_2Av = c_1v + c_2Av = 0$, thus $\underline{c_2 = 0}$.

- (b) Set $P = \begin{pmatrix} Av & v \end{pmatrix}$, then P is invertible by (a). Moreover,

$$AP = \begin{pmatrix} A^2v & Av \end{pmatrix} = \begin{pmatrix} 0 & Av \end{pmatrix} = P \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

- (c) By (b),

$$P^{-1}A^2P = (P^{-1}AP)^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^2 = 0_{2 \times 2}$$

and hence $A^2 = 0_{2 \times 2}$.

4. (a) $\text{tr}(PQ) = \sum_{k=1}^n (PQ)_{kk} = \sum_{k=1}^n \left(\sum_{j=1}^n p_{kj} q_{jk} \right) = \sum_{j=1}^n \left(\sum_{k=1}^n q_{jk} p_{kj} \right) = \sum_{j=1}^n (QP)_{jj} = \text{tr}(QP)$.

- (b) (i) Proof: By the assumption, $A = Q^{-1}BQ$ for some invertible matrix Q . Using (a),

$$\text{tr}(A) = \text{tr}(Q^{-1}BQ) = \text{tr}(QQ^{-1}B) = \text{tr}(B).$$

- (ii) Disproof: Take $n = 2$, $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, then $\text{tr}(A) = 0 = \text{tr}(B)$, but A and B are not similar because

$$Q^{-1}BQ = 0_{2 \times 2} \neq A$$

for any invertible matrix Q .